CS 352 Project 2: Domain Name System

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Overview

For this project we wrote a DNS resolver in Python in three parts. Part A sends recursive A and AAAA queries to Google's public DNS server (8.8.8.8). Part B adds support for NS records. Part C does iterative resolution, starting at a root server and following referrals until it reaches the authoritative server. We used Wireshark to look at the packets for each part, and the answers below are based on those captures.

Part	What it does	Test we ran
A	Recursive lookup	A and AAAA queries to 8.8.8.8
B	NS records	NS query for princeton.edu (9 answers)
C	Iterative lookup	ilab1.cs.rutgers.edu: root, then .edu, then Rutgers

Part A: Recursive Resolver

We sent queries to 8.8.8.8 with RD=1, so the server did the full lookup for us. Unless noted, the examples use the A record query for ilab1.cs.rutgers.edu.

1. What is the domain name in the question section (hex)?

`05 69 6C 61 62 31 02 63 73 07 72 75 74 67 65 72 73 03 65 64 75 00`

Each label starts with a length byte. 05 is followed by "ilab1", 02 by "cs", 07 by "rutgers", and 03 by "edu". The 00 at the end marks the end of the name.

2. What is the domain name in the answer section (hex)? Explain.

`C0 0C`

This is a compression pointer instead of the full name. It points to offset 0x000C (12), which is where the name starts in the question section, right after the 12-byte header. This way the server doesn't have to repeat the whole name.

3. What is RDLENGTH in the response (hex)?

`00 04`. An A record holds an IPv4 address, which is 4 bytes.

4. What address was returned (hex)?

`80 06 0D 02`, which is 128.6.13.2, the IP address of ilab1.cs.rutgers.edu.

5. Why is ANCOUNT 3 for www.princeton.edu when only two IP addresses are returned?

ANCOUNT counts all the records in the Answer section, not just addresses. The response has one CNAME record (www.princeton.edu is an alias) and two A records, so the total is 3.

Part B: NS Records

We sent an NS query for princeton.edu to 8.8.8.8 with RD=1. The response had nine name servers in the Answer section.

6. What are QDCOUNT, ANCOUNT, NSCOUNT, and ARCOUNT (hex)?

QDCOUNT = `0x0001`, ANCOUNT = `0x0009`, NSCOUNT = `0x0000`, ARCOUNT = `0x0000`.

So the response had one question, nine answers, and nothing in the Authority or Additional sections.

7. What are the RDLENGTH values for all resource records (hex)?

From the nine NS records in our capture:

Name server	RDLENGTH
ns6.dnsmadeeasy.com	`0x0015`
ns5.dnsmadeeasy.com	`0x0006`
a24-66.akam.net	`0x0011`
a3-67.akam.net	`0x0008`
ns7.dnsmadeeasy.com	`0x0006`
a7-65.akam.net	`0x0008`
a20-65.akam.net	`0x0009`
a6-64.akam.net	`0x0008`
a1-158.akam.net	`0x0009`

8. What name servers were returned (hex)?

These are the NS RDATA bytes from the princeton.edu response (Frame 22). C0 2F and C0 65 are compression pointers to the dnsmadeeasy.com and akam.net suffixes written earlier in the same message.

Name server	NS RDATA (hex)
ns6.dnsmadeeasy.com	`03 6E 73 36 0B 64 6E 73 6D 61 64 65 65 61 73 79 03 63 6F 6D 00`
ns5.dnsmadeeasy.com	`03 6E 73 35 C0 2F`
a24-66.akam.net	`06 61 32 34 2D 36 36 04 61 6B 61 6D 03 6E 65 74 00`
a3-67.akam.net	`05 61 33 2D 36 37 C0 65`
ns7.dnsmadeeasy.com	`03 6E 73 37 C0 2F`
a7-65.akam.net	`05 61 37 2D 36 35 C0 65`
a20-65.akam.net	`06 61 32 30 2D 36 35 C0 65`
a6-64.akam.net	`05 61 36 2D 36 34 C0 65`
a1-158.akam.net	`06 61 31 2D 31 35 38 C0 65`

9. Why do NS records have different RDLENGTH values?

The data in an NS record is a domain name, so the length depends on the name. The first time a suffix like dnsmadeeasy.com or akam.net appears, it is written out in full. After that, the server replaces it with a 2-byte pointer. For example, ns6.dnsmadeeasy.com is 21 bytes (0x15), but ns5.dnsmadeeasy.com is only 6: one length byte, "ns5", and the pointer C0 2F.

10. Why are the NS records in the Answer section and not Authority?

We asked for NS records (QTYPE=NS), so they are the answer to our question. The Authority section is used when a server is referring you to another server, like the root and .edu servers do in Part C.

Part C: Iterative Resolver

Here we set RD=0 and started at the root server 199.9.14.201. The resolver followed UDP referrals to a .edu server and then to the Rutgers authoritative server. None of the three responses in the final capture were truncated, so no retry was needed.

Step	Server	Result
Root	`199.9.14.201`	Referral to .edu
.edu	`192.5.6.30`	Referral to rutgers.edu
Rutgers	`130.156.133.30`	A record: 128.6.13.2

11. Why is TC=1 in the first root server response?

In an earlier test using root server 198.41.0.4, the UDP response had TC=1, meaning the full response did not fit in the UDP message. As the project requires, our updated resolver stops and reports an error when it gets TC=1. Our final capture uses root server 199.9.14.201, whose response was not truncated (TC=0), so the TC=1 response does not appear in it.

12. Which of the three responses has AA=1, and why are the others AA=0?

The third response, from 130.156.133.30, has AA=1 because that server is authoritative for rutgers.edu and gave the actual A record for ilab1.cs.rutgers.edu. The root and .edu servers only sent referrals to the next server, so they have AA=0.

13. What is the TYPE field in the first iterative query (hex)?

`0x0001` (A), since we were asking for the IPv4 address of ilab1.cs.rutgers.edu.

14. What TYPE values are in the first iterative response?

NS (`0x0002`) records in the Authority section for the .edu servers, and A (`0x0001`) and AAAA (`0x001C`) glue records in the Additional section with their addresses. Unlike the earlier test, this root response came back complete over UDP.

15. What is the destination IP of the second DNS query (hex), and where did it come from?

It goes to the .edu server at 192.5.6.30, which is `C0 05 06 1E` in hex. The resolver got this address from an A glue record in the Additional section of the root server's referral.

Conclusion

The final Part C capture has six DNS packets for one lookup: three queries and three responses, all with transaction ID 0x0539 (1337). The resolver went from the root server 199.9.14.201 to the .edu server 192.5.6.30 and then to the Rutgers server 130.156.133.30. The final response returned 128.6.13.2 with AA=1, RD=0, RA=0, TC=0, and RCODE=0.

Doing the three parts showed the difference between using a recursive DNS server, asking for NS records directly, and following referrals ourselves. The Authority section tells you which name servers a zone is delegated to, and the Additional section gives their glue IP addresses. The program also checks the TC flag and exits with an error if a response is truncated, as the assignment requires.

Submission Notes

The code and packet captures for Parts A, B, and C are in our repository. The Part C answers match the final UDP capture. The TC=1 response mentioned in Question 11 is from an earlier separate test, not the final capture.

Repository: github.com/ethanschivel/cs352-project2